Self-Supervised Learning Guided by SAR Image Factors for Terrain Classification

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***Abstract*— Effective feature representation is the key to synthetic aperture radar (SAR) image terrain classification. Limited by the abstract appearance and the scarcity of high-quality labeled data in this field, the features learned by current methods, especially deep learning models, do not have enough directivity and applicability, which hampers the performance. This article proposes multi-image factor self-supervised learning (MFSSL) to achieve directional feature learning and obtain generalized features with few patch-level labeled data. The framework consists of an upstream multifactor image style transfer task and a downstream terrain classification task. In the upstream task, the goal of feature learning is first set up by multiple SAR image factors, including the observation region, the terrain category, and the imaging parameters. Then, different styles of SAR terrain images are generated and reconstructed under this goal. Through this bidirectional generative learning, the low-level external appearance of the terrain is removed, while the essential and discriminative feature representation is retained and shared across different factors. Finally, the downstream model inherits the general feature from the upstream model and implements the terrain classification task using a small amount of labeled data. Experiments conducted on three broad SAR scenes with different image factors demonstrate that the proposed framework can improve pixel-level terrain classification only with a few patch-level labeled data.**

***Index Terms*— Image factors, self-supervised learning (SSL), synthetic aperture radar (SAR), terrain classification.**

I. Introduction

DUE to active detecting and microwave properties, synthetic aperture radar (SAR) can penetrate fog, clouds, rain, tree crowns, and even the Earth's surface to capture information about objects. This unique characteristic makes SAR a powerful observing tool for civil and military applications. SAR terrain classification is essential to SAR understanding and interpreting, such as target recognition and change detection. More directly, it makes excellent contributions to land cover [1], polar monitoring [2], resource monitoring [3], geographical mapping [4], and so on.

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The general SAR image terrain classification methods include data preprocessing, feature extraction, and classifier design. Among them, feature extraction is the key to the performance of terrain classification and draws the most attention. In recent decades, countless SAR image feature representation methods have been proposed, from the early popular traditional descriptors to the current hottest deep-learning-based methods [5]. In traditional methods, the extracted feature is explicitly determined by the descriptors. For example, the GLCM [6], [7] statistics describe pixel pairs' spatial distribution patterns. The methods [8], [9] based on the wavelet transform classify the terrain according to its characteristics in the frequency domain. Assuming terrain features exist within a linear subspace, the terrain is described using sparse representation [10], [11] with a vector of sparse projection coefficients. However, the feature extracted from the above traditional descriptors is limited by the biased designs and the complex hyperparameter settings. Owing to the automatic and end-to-end feature learning paradigm, deep-learning-based methods dominate recent research. CNN is usually used to obtain high-level deep semantic features from SAR scenes [12]. LSTM is utilized to capture complementary information among different feature maps and correlations between elements of the coherence matrix [13]. ViT [14], [15] can remedy the defects of CNN in capturing global information of images. In general, although traditional and deep-based methods have demonstrated effectiveness depending on intricate operations or network architectures, they often overlook the analysis of SAR images at the feature level, impeding significant advancements in terrain classification performance.

Data are always the driving force for the algorithm, especially for data-driven deep learning algorithms. Chen et al. [16] show that the classification performance of the model improves rapidly as the amount of training data increases under a specific range. However, facing limited labeled data, overfitting can easily happen in deep learning models, resulting in less effective feature learning. As for SAR terrain classification, the risk of overfitting is more salient due to the complexity of the terrain and imaging process. Specifically, on the one hand, the variety of terrain displayed in the broad scenes and the limitation of data lead to a relatively small number of samples under a specific region [17]. On the one hand, different from natural images, which are highly consistent with human visual cognition, SAR images display many abstract forms such as foreshortening, shadow, and layover because of the active microwave imaging process. This requires a high professional

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level for data labeling, making high-quality annotation scarce [18]. In summary, designing algorithms that are less dependent on labeled data is an urgent practical requirement to alleviate the above dilemma.

Limited by the quantity and complexity of SAR image data, most current SAR classification methods are tailored for training and testing data from the same imaging parameters or scenes. In the research of Zhao et al. [19], [20], though the training sets of scene classification are conducted from different areas, they are all from the same sensors and bands (TerraSAR-X satellite with an HH-polarization spotlight mode). In our previous research [21], [22], the proposed models are all training and testing on the same broad scenes though the proportion of the picked training data is as small as possible. These scant data-based models hinder the generalization of various SAR scenes in real, and one way to relieve this obstacle is transfer learning. Tai et al. [23] proposed a few-shot transfer learning strategy based on Bayesian convolutional neural networks. Transferring the knowledge from the electro-optical domain to the SAR domain, the dependence of the model on additional SAR samples is alleviated. Zhao et al. [24] proposed a spherical space domain adaptation network under open set conditions to distinguish five ship classes across Sentinel-1 and Gaofen-3. Nevertheless, such methods are still in the early stages, affected by huge differences between transfer domains. More targeted transfer methods need to be proposed to reduce the gap and enhance the generalization.

From the above description, current supervised SAR terrain classification methods exhibit ambiguous learning directions, a firm reliance on labeled data, and a limited scope of application. These factors contribute to inadequate feature learning and low generalization capabilities. As a learning strategy with low labeling requirements and high generalization ability, self-supervised learning (SSL) can adopt self-defined pseudolabels to supervise upstream tasks and use the learned representations for downstream tasks. Moreover, in SAR terrain classification, SSL exhibits distinctive advantages. First, SSL excavates feature representations from complex and diverse SAR data and has robust domain adaptability. Second, SSL leverages the abundant SAR data without necessitating expensive annotations. Consequently, it has garnered extensive attention and research focus in natural and remote sensing domains [25], [26], [27]. As the mainstream method of SSL, generative SSL acquires feature representations by generating or reconstructing either partial or entire input images [28], [29], [30], [31]. Wang et al. [15] used the mask autoencoder method of the ViT structure to implement the land cover classification of PolSAR images. Sun et al. [32] proposed a remote sensing basic model framework based on generative SSL to encode image content and reconstruct the mask area under the patch incomplete mask strategy. The investigation above delves into the viability of SSL in terrain classification. Nevertheless, these features are contingent on the guidance of pretext tasks and still exhibit biases in specific tasks, lacking a clear learning direction. It is worth noting that the SAR active imaging process relies not solely on the intrinsic characteristics of the imaging target but is also intricately linked to the observing system, band, polarization mode, and so on. To simplify the expression in this article, we denote the inherent information of the target in the SAR image as content features and the additional information influenced by the observing system (radar factors) as style features. Therefore, we characterize the extrinsic factors influencing the stylistic representation of SAR images as SAR image factors.

The above research confirms the feasibility of SSL in remote sensing image interpretation. In addition, the image style transfer task [33], [34] can transfer the style of the input image content into another style, exhibiting powerful feature learning, which can be a good choice for the upstream task. Inheriting the advantages of SSL and profoundly considering the characteristics of endogenous (terrain) and exogenous (style) information of SAR images, a multi-image factor SSL (MFSSL) framework is proposed for large-scene high-resolution SAR image terrain classification. Our attention is directed toward elucidating the connection between multifactors and SAR image characteristics. We explicitly integrate multifactors into the upstream task, utilizing cost-effective, patch-level data. The objective is to instruct the model to eliminate style attributes influenced by multifactors and to capture common representations of content features, thereby enhancing the discriminative and generalizable aspects of image features. MFSSL only implements pixel-level terrain classification in the downstream task using a small amount of labeled data.

The main contributions of this article are given as follows.

1) A generative self-supervised framework for SAR image classification framework is proposed, which explicitly merges various SAR image factors and gives clear direction for feature learning, improving the discrimination of the learned feature.
2) MFSSL can automatically capture the image feature representation in the upstream task and effectively transfer it to the downstream task, achieving superior performance than supervised methods under a few low-cost data.
3) With a single encoder extracting image features under the guidance of different image factors in the upstream task, the desirable results on multiple disparate high-resolution SAR scenes are obtained by the fine-tuned downstream network, which convincingly demonstrates the generalization of MFSSL.

The structure of the remainder of this article is given as follows. Section II details the generative SSL algorithm. Section III presents the experimental results and the corresponding analysis. Section IV gives the conclusion and future direction.

II. Method

This section elaborates on the technical process of MFSSL. A complete framework is shown in Fig. 1, in which MFSSL consists of two stages: the upstream task for SAR image style transfer learning and the downstream task for SAR image terrain classification. In the upstream task, domain information explicitly includes the imaging area and radar parameters, and mutual style transfer learning is continued.

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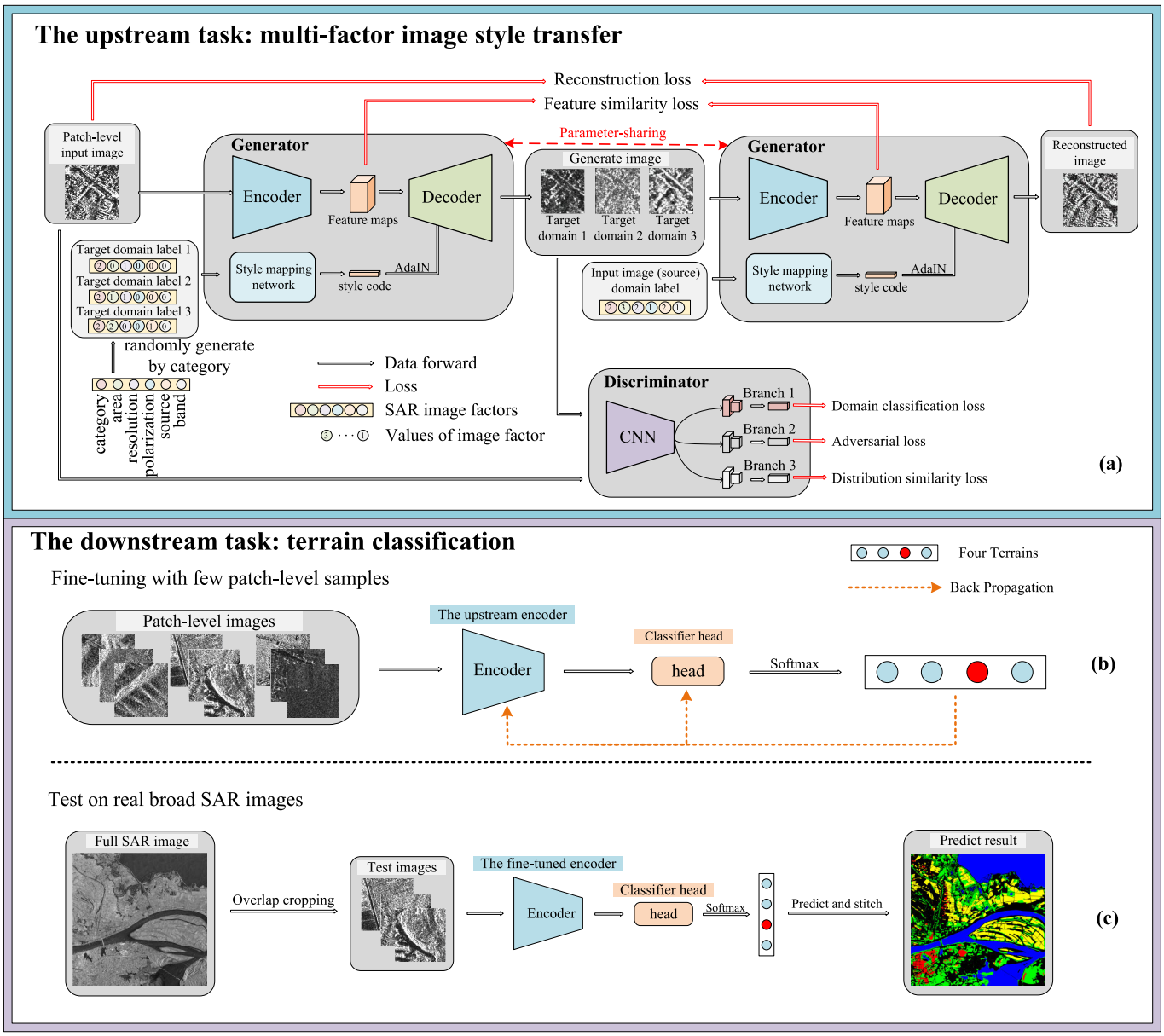


Fig. 1. Illustration of MFSSL for training the upstream model, fine-tuning, and testing on the downstream terrain classification task. Given a patch-level image, the generator is guided by randomly generated domain labels to produce target-style images in an adversarial manner, making them indistinguishable. After pretraining, we attach a classification head to the encoder and fine-tune the overall parameters. The specific parameters of each module in the framework are shown in Tables II–V and VII. (a) Pipline of multifactor image transfer task. (b) Terrain classification training process. (c) Terrain classification testing process.

This approach allows the extraction of feature representations from multidomain SAR images using a single encoder, thereby enhancing generalization capabilities. The pretrained encoder is transferred and connected to the classifier in the downstream task. All classifier parameters are fine-tuned with a small amount of low-cost patch-level label data. This fine-tuning allows MFSSL to alleviate the need for labeled data largely. In the test part, the well-trained network is used to predict the category of each pixel in the large scenes.

Section II-A introduces low-cost patch-level training samples obtained from unlabeled large-scene high-resolution SAR images and their corresponding domain labels. Section II-B describes the multifactor image style transfer upstream task. In Section III-C, the downstream task of terrain classification is presented.

A. Generation of Training Data and Domain Labels

This section details the construction of the dataset and labels. Specifically, for broad SAR scenes, the actual land surface coverage is extensive, and each class of terrain always covers a large area, such as large lakes, patches of forests, buildings, and farmland. As shown in Fig. 2, comparing the SAR image with its corresponding optical image, it is easy to find rich regions of the above four terrains in the SAR image (marked by boxes with different colors). A patch-level training dataset is constructed by sampling from these regions. Compared with pixel-level manual labeling, this artificial-assistant sampling does not require a high-level professional background in SAR. The acquisition cost is negligible, and the obtained dataset satisfies the training of MFSSL.

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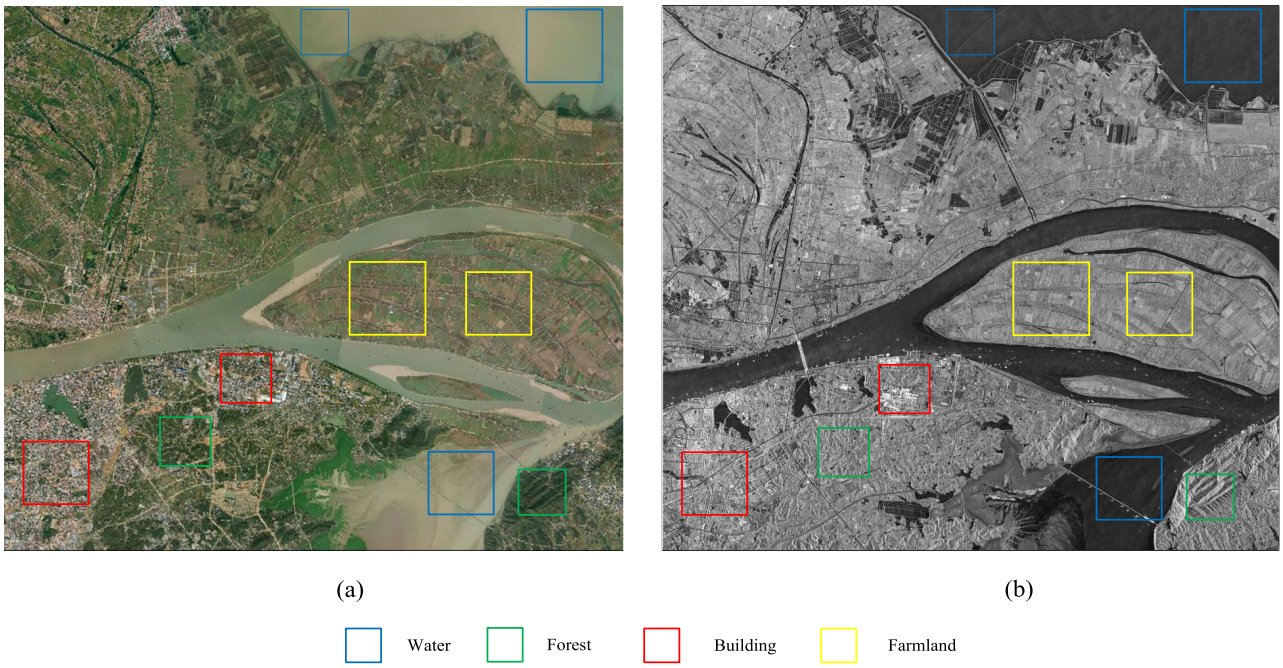


Fig. 2. Construction of patch-level training set from Jiujiang (size: 7357 × 8000). The regions of the same terrain category in the unlabeled SAR image and the corresponding optical image are shown in the same color. (a) Optical image. (b) SAR image.

The above process determines the patches and their corresponding categories. The imaging area, resolution, polarization, band, and sensor correspond to the terrain distribution, detail, brightness, hue, and imaging mode information in the SAR image, respectively. These image factors and categories comprise the training samples' domain labels. Note that the category is included as a component of the domain label. The value of the category component cannot be modified arbitrarily, unlike other elements, as it serves as a constraint. Moreover, the source-domain and generated images must belong to the same category. This choice is grounded in two considerations. First, substantial variations exist among terrain, encompassing size, shape, and physical properties. Employing categories as constraints in the domain label streamlines the problem and prevents biases toward other terrain. Second, it facilitates the exploration of the shared representation of SAR images within the same category. Thus, the domain label is a 6-D vector. Each element in this vector represents the specific attribute value of the corresponding image factor. Theoretically, any existing attribute value can be chosen randomly. However, considering the guidance requirement by real data and the quantitative measurement, the attribute values are randomly selected among our data, as listed in Table I.

B. Upstream Multifactor Image Transfer Task

The upstream multifactor image style transfer network is an image generation model based on GAN, which flexibly transforms the source image to the target-domain image according to the given domain label from the target domain. It achieves mutual conversion between styles of SAR images under different image factors. The GAN-based training framework encodes the intrinsic content features of SAR images, explores shared representations among images that depict the same terrain, and eliminates extraneous style features influenced by multifactor. Meanwhile, the decoder combines multifactor style codes from the target domain with content features of the image to generate falsifiable images. In essence, the multifactors guide the model to mitigate the impact of specified exogenous factors on content features in unlabeled SAR images from various scenes. It produces feature representations shared among the same terrain, offering a generalized pretrained encoder for the downstream terrain classification task and alleviating the dependence on labeled data.

The structure of the upstream multifactor image style transfer model is shown in Fig. 1(a). Adapted from StarGAN v2 [35], the model contains a parameter-sharing generator G and a discriminator D, where G consists of an encoder E, a decoder H, and a style mapping network M. The specific structures of the above modules are shown from Tables II to V, respectively.

G transfers the source-domain image s to the target domain under the specified style by the target-domain label. Concretely, given the input s and target-domain label l_t, the encoder E obtains feature maps $f_s = E(s)$ from the source-domain image. l_t is acquired by randomly sampling the domain labels l_s from the batch input data within the same category, ensuring a real image guide for the target domain. At the same time, the style mapping network M maps the specified target-domain labels to high-dimensional style codes $c_t = M(l_t)$ to expose potential relationships among various image factors and enable fine control over styles. The adaptive instance normalization (AdaIN) [33] is adopted to inject c_t into the decoder H and control the style of the generated image $s' = H(f_s, c_t)$. In this process, the attribute value of the category in l_t is fixed for the input image and its corresponding output image to avoid the confusion of domain and semantic information caused by excessive style differences between categories. It ensures the interpretability of image features.

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It is noteworthy that different l_t's still have different category attribute values. The generator G can learn the features of different terrains. The mapping relationship between the source- and target-domain images is captured under explicit guidance and strengthens the feature representation.

For discriminator D, except for the common source distinguish, an auxiliary domain classifier and a distribution feature extractor are added to identify the domain attributes and measure the distribution difference between s and s'. For a given input image x, D produces probability distributions of the source and domain attribute and outputs image distributional features, $D : x \rightarrow \{D_{\text{src}}(x), D_{\text{cls}}(x), D_{\text{dis}}(x)\}$. These auxiliary branches also provide additional gradient information to the generator, boosting the generalization ability of the generator under different styles.

In summary, the loss function of MFSSL consists of the following parts: an adversarial loss L_{adv}, a domain classification loss L_{cls}, a feature similarity loss L_{fs}, a reconstruction loss L_{rec}, and a distribution similarity loss L_{dis}. For stable training and generating high-quality images, the objective of Wasserstein GAN [36] with gradient penalty is taken as the adversarial loss, which is defined as follows:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}[D_{\text{src}}(s)] - \mathbb{E}\left[D_{\text{src}}(H(f_s, c_t))\right] - \lambda_{\text{gp}}\mathbb{E}\left[\left(||\nabla_{\hat{s}} D_{\text{src}}(\hat{s})||_2 - 1\right)^2\right] \tag{1}$$

where $\hat{s}$ is sampled uniformly along a straight line between a pair of source- and target-domain generated images and λ_{gp} is the gradient penalty weight coefficient.

For the multidomain image style transfer task, the generator produces target images with different styles, while the discriminator learns to distinguish between them. Thus, the domain classification loss is defined as follows:

$$\mathcal{L}_{\text{cls}}^{\text{real}} = \sum_{i=1}^{C} \mathbb{E}\left[-\log D_{\text{cls}}\left(l_s^i | s\right)\right] \tag{2}$$

$$\mathcal{L}_{\text{cls}}^{\text{fake}} = \sum_{i=1}^{C} \mathbb{E}\left[-\log D_{\text{cls}}\left(l_t^i | s'\right)\right] \tag{3}$$

where $C = 6$ is the number of domain attributes and l_s^i and l_t^i represent the values of the ith attribute of l_s and l_t, respectively. During the interactive training, the generator minimizes the domain classification loss of fake images to generate target-domain images with plausible styles. Meanwhile, the discriminator improves its ability to discriminate.

Adversarial and domain classification losses aid in producing target-style images but do not ensure content consistency. In other words, the feature information of the source image may be lost. Therefore, the generated image s' is also used as the input by the parameter-sharing generator to reconstruct a plausible real image. A cycle consistency loss [37] is adapted to constrain the content consistency between the reconstructed and real images. Simultaneously, the loss for feature similarity is incorporated to minimize the difference in feature maps between the source and generated images. The above two losses can simultaneously draw the real and generated images close with the feature and domain style information, increasing the generality of E on SAR images with different radar parameters. The reconstruction loss and the feature similarity loss are defined as follows:

$$\mathcal{L}_{\text{rec}} = \mathbb{E}\left[||s - H\left(E\left(s'\right), M(l_s)\right)||_1\right] \tag{4}$$

$$\mathcal{L}_{\text{fs}} = \mathbb{E}\left[||E(s) - E\left(s'\right)||_2^2\right]. \tag{5}$$

In addition, the overall generation ability is improved by utilizing an image distribution similarity to draw the distribution distance of the generated image closer to that of the real image, which is defined as

$$\mathcal{L}_{\text{dis}} = \mathbb{E}\left[\text{KL}\left(D_{\text{dis}}(s) - D_{\text{dis}}\left(s'\right)\right)\right] \tag{6}$$

where KL divergence [38] is used to measure the similarity of distribution.

To sum up, the objective functions of G and D are defined as follows:

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}}^{\text{fake}} + \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{sim}}\mathcal{L}_{\text{fs}} + \lambda_{\text{dis}}\mathcal{L}_{\text{dis}} \tag{7}$$

$$\mathcal{L}_D = -\mathcal{L}_{\text{adv}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}}^{\text{real}} \tag{8}$$

where $\mathcal{L}_G$ is the objective function of G and $\mathcal{L}_D$ is that of D. λ_{cls}, λ_{rec}, λ_{fs}, and λ_{dis} represent the weight coefficients of domain classification loss, reconstruction loss, feature similarity loss, and distributional similarity loss, respectively.

C. Downstream Terrain Classification Task

The flowchart of the downstream terrain classification task is shown in Fig. 1(b) and (c). The pretrained model for the downstream terrain classification task is the encoder E from the upstream task. Based on this, the downstream task involves fine-tuning with a few patch-level label samples and prediction on broad SAR scenes, respectively, described as training and testing processes in the downstream task.

The downstream classification model P consists of an encoder and a classifier head, where the encoder is taken directly from the upstream multifactor image style network. We fine-tuned the P using a small amount of patch-level labeled data obtained in Section II-A to improve the performance under supervision and avoid the dilemma of insufficient labeled data. In this phase, the output of the model is the probability distribution of the terrain, and the multiclass cross-entropy loss is used as the objective function, which is defined as follows:

$$\mathcal{L}_p = \mathbb{E}\left[-\log P(l_x | x)\right] \tag{9}$$

where x is the patch-level training sample and l_x is the corresponding terrain category. The whole training process of MFSSL is listed in Algorithm 1.

In the predicted phase, the broad unlabeled high-resolution SAR scenes are sliced into overlapping blocks as the downstream model input P. Then, the central region of the predicted block is selected and spliced to achieve pixel-level terrain classification. In practice, the number of fine-tuning samples, the cropped block sizes, and the selected central region sizes are essential to the performance, whose determination is given in Section III-C2.

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Algorithm 1 Training Process of MFSSL

Input:
source domain images s, source domain labels l_s, and target domain labels l_t, the encoder E, the style mapping network M, the decoder H, the discriminator D, and the classification model P.

Output:
the classification model P.

The upstream task:
Generate samples and the target domain labels l_t with the same terrain category.
for number of iterations **do**
  Calculate the source domain image features $f_s = E(s)$ and the target domain style code $c_t = M(l_t)$.
  Obtain target domain style images $s' = H(f_s, c_t)$.
  Feed s and s' to the discriminator D, and calculate the objective function of D by formula 8.
  Update the weights of the discriminator D.
  Obtain generated image feature $E(s')$ and reconstructed image $H(E(s'), M(l_s))$, then calculate the losses by formula 5 and formula 6.
  Feed s and s' to the discriminator D, and calculate the objective function of the generator by formula 7.
  Update the weights of E, M, and H.
end for
In the downstream task:
Select a few samples of each class from x and their corresponding category labels l_x as the training set.
Transfer the weights from E to P.
for epochs **do**
  Obtain predicted probability distribution $P(x)$, and calculate classification loss by formula 9.
  Finetune the overall weights in the classification model P.
end for

III. Experiments

This section presents the experimental results and analysis of MFSSL. Section III-A introduces the data and experimental conditions. The performance of multifactor image style transfer in the upstream task is shown in Section III-B. The results of the downstream terrain classification are shown in Section III-C. The ablation study is performed in Section III-D.

A. Experimental Data and Setting

Four high-resolution broad SAR scenes are selected to conduct the following experiments, and all are imaged in the Northern and Eastern Hemispheres. They contain multiple image factors and levels of domain difference information. Therefore, they are all applied to upstream tasks to provide rich guidance information. Napoli and PoDelta, captured by COSMO-SkyMed, offer a regional attribute comparison under identical radar factors. Rosenheim, imaged by TerraSAR-X, further enhances the distinction between the scenes above regarding resolution. Jiujiang, captured by GF-3, broadens the differentiation to encompass various factors. Moreover, in the terrain classification task, due to the limited scene coverage of the Rosenheim data, we only validate the effectiveness and generalization of the proposed method on the remaining three extensive scene data. The raw information of SAR image is stored in 16 bit, while it is not fit for the standard 8-bit image processing. Therefore, all raw SAR images map pixel values to $(0, 255)$ using truncated linear stretch [39]. More detailed data information is shown in Table I. The four broad scenes contain the same four terrain categories, i.e., water, farmland, forest, and building, and their color indicator is illustrated at the bottom of predicted results. Also, the category of others is neglected during training and testing. All experiments in this article are implemented on the framework of PyTorch 1.7.1 with an NVIDIA RTX 2080Ti GPU.

B. Multifactor Image Style Transfer in the Upstream Task

1) Settings and Metrics: The training dataset consists of patch-level labeled samples from the method proposed in Section II-A. Specifically, in each SAR scene, 400 samples with a size of 200×200 are picked for each category. The domain label of each block is subsequently determined according to the data information in Table I, whose dimension is 6.

In the upstream task, the generator G includes the encoder E, the style mapping network M, and the decoder H, whose structures are shown in Tables II, III, and IV, respectively. The structure of E and H is approximately symmetrical except for the normalization layer. In order to fuse the style code with M, we replaced the normalization layer of H with the AdaIN layer. The structure of discriminator D is shown in Table V. Three convolutional layers compose the output branches: the adversarial discrimination module, the class discrimination module, and the distribution feature module, respectively. λ_{cls}, λ_{rec}, λ_{sim}, λ_{dis}, and λ_{gp} are picked as 1, 50, 50, 0.1, and 20, respectively, in the experiment. The batch size is 32 here. The number of iterations is 100 000. The initial learning rate of the upstream model is 0.0001, and the learning rate decay strategy of step descent is adopted. The Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is used for model training. The selection of the above parameters mainly refers to StarGAN [34].

Eight metrics are selected to evaluate the quality of the generated images in the upstream multifactor image style transfer tasks, including mean, variance, dynamic range (DR), equivalent numbers of looks (ENL), luminance similarity (LS), contrast similarity (CS), 1-NN classifier accuracy (1NN-ACC), and the Fréchet inception distance (FID) [40], which are defined as follows.

Mean (μ_x): It reflects the average backscattering intensity of the terrain under different regions and imaging parameters

$$\mu_x = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} I_{i,j} \tag{10}$$

where H and W represent the height and width of the image, respectively, and $I_{i,j}$ represents the gray-scale value of the image at pixel (i, j).

Variance (σ_x^2): It describes the degree of deviation of image pixel values from the mean and the inhomogeneity of the

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TABLE I
INFORMATION OF SAR SCENES

Area	Time	Sensor	Image Sizes	Resolution	Band	Polarization	Category
Napoli	2013-03-01	Cosmo-SkyMed	18332 × 16000	2.5m	X	HH	Water
PoDelta	2007-09-27	Cosmo-SkyMed	18308 × 16716	2.5m	X	HH	Forest
Rosenheim	2008-01-27	TerraSAR-X	7691 × 7224	1.75m	X	HH	Building
JiuJiang	2016-11-24	GF-3	8000 × 8000	3m	C	DV	Farmland

TABLE II
ARCHITECTURE OF THE ENCODER

Layer	Resample	Norm	Output shape
Image	-	-	128 × 128 × 3
Conv 1×1	-	-	128 × 128 × 32
ResBlk	AvgPool	IN	64 × 64 × 64
ResBlk	AvgPool	IN	32 × 32 × 128
ResBlk	AvgPool	IN	16 × 16 × 256
ResBlk	AvgPool	IN	8 × 8 × 512
ResBlk	-	IN	8 × 8 × 512
ResBlk	-	IN	8 × 8 × 512

*Number of parameters : 1.197×10^7.

TABLE III
ARCHITECTURE OF THE STYLE MAPPING NETWORK

Layer	Actvation	Output shape
Domain label	-	6
Linear	ReLU	512
Linear	ReLU	512
Linear	ReLU	512
Linear	ReLU	512
Linear	ReLU	512
Linear	ReLU	512
Linear	ReLU	512
Linear	ReLU	64

*Number of parameters : 1.612×10^6.

TABLE IV
ARCHITECTURE OF THE DECODER

Layer	Resample	Norm	Output shape
Feature	-	-	8 × 8 × 512
Style code	-	-	64
ResBlk	-	AdaIN	8 × 8 × 512
ResBlk	-	AdaIN	8 × 8 × 512
ResBlk	Upsample	AdaIN	16 × 16 × 256
ResBlk	Upsample	AdaIN	32 × 32 × 128
ResBlk	Upsample	AdaIN	64 × 64 × 64
ResBlk	Upsample	AdaIN	128 × 128 × 32
Conv1×1	-	-	128 × 128 × 3

*Number of parameters : 1.241×10^7.

TABLE V
ARCHITECTURE OF THE DISCRIMINATOR D

Layer	Resample	Norm	Output shape
Image	-	-	128 × 128 × 3
Conv3×3	-	-	128 × 128 × 16
ResBlk	AvgPool	-	64 × 64 × 32
ResBlk	AvgPool	-	32 × 32 × 64
ResBlk	AvgPool	-	16 × 16 × 128
ResBlk	AvgPool	-	8 × 8 × 256
ResBlk	AvgPool	-	4 × 4 × 512
Branch 1:Conv3×3	-	-	4 × 4 × 1
Branch 2:Conv4×4	-	-	1 × 1 × 18
Branch 3:Conv4×4	-	-	1 × 1 × 128

*Number of parameters : 3.734×10^6.

style-transferred image, which can effectively represent the details of images

$$\sigma_x^2 = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left(I_{i,j} - \mu_x\right)^2 \tag{11}$$

where μ_x represents the mean of the image.

DR: It describes the difference between the maximum and minimum values of all pixel values in an image

$$\mathrm{DR} = 10 \lg \left(\frac{I_{\max}}{I_{\min}} \right) \tag{12}$$

where $I_{\max}$ and $I_{\min}$ represent the image's maximum and minimum pixel values, respectively. The dynamic range is large for the terrain categories with various backscatters, such as forests and buildings. The dynamic range is small for the terrain categories with a single backscatter, such as water and farmland.

ENL: It reflects the intensity of image noise

$$\mathrm{ENL} = \frac{\mu_x^2}{\sigma_x^2}. \tag{13}$$

LS and CS: They describe the similarity of brightness and contrast between generated and real images. The definitions are consistent with brightness similarity and contrast similarity in SSIM [41]

$$\mathrm{LS} = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \tag{14}$$

$$\mathrm{CS} = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \tag{15}$$

where $C_1 = (k_1 L)^2$ and $C_2 = (k_2 L)^2$. L is the dynamic range of pixel values, $k_1 = 0.01$, and $k_2 = 0.03$.

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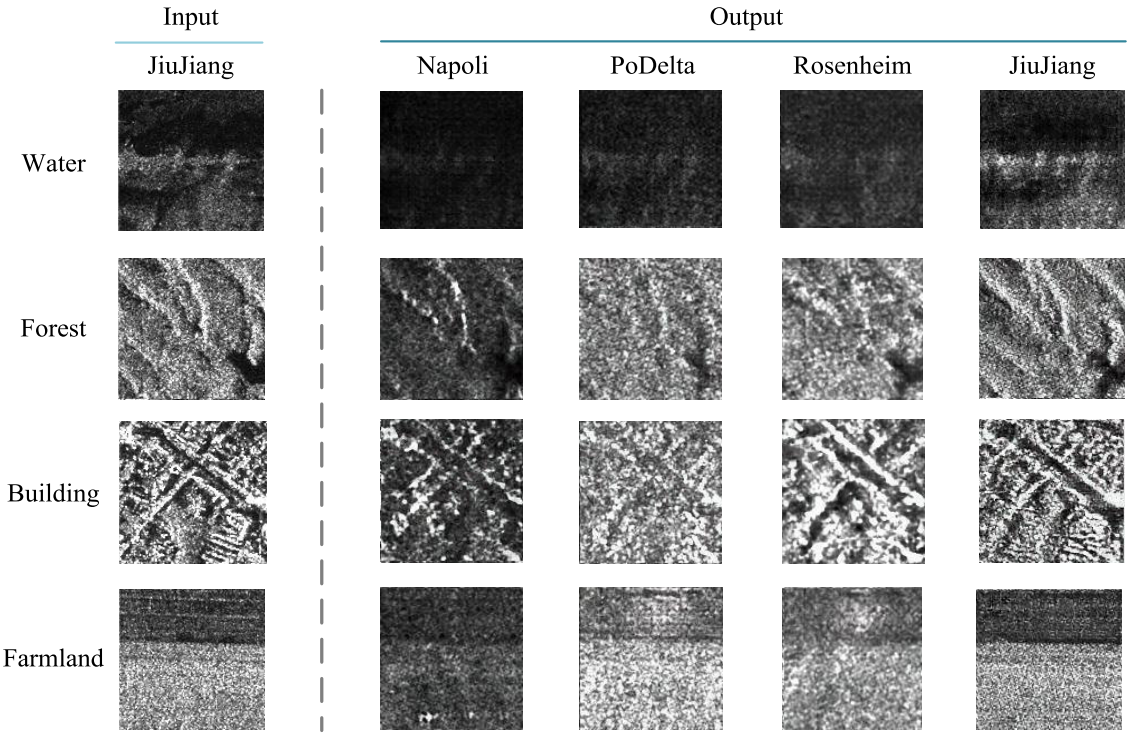


Fig. 3. Multifactor image style transfer results from Jiujiang to four broad scenes. The first column is the real image of each terrain category, and the rest are generated or reconstructed images under different target-domain styles.

1NN-ACC: It describes the performance of the image generated by the model. When the accuracy is 0, the model cannot effectively generate images. When the accuracy is 1, the model only performs a simple memory. When the accuracy is 0.5, the model has the best generation performance

$$1\text{NN}-\text{ACC}=\frac{1}{n_1+n_2}\sum_{i=1}^{n_1+n_2}A\left(l_c^i=l_y^i\right) \tag{16}$$

where n_1 and n_2 represent the number of real samples and generated samples, respectively, l_c^i indicates the real terrain category label of the ith sample, and l_y^i indicates the terrain category of the sample with the closest L_2 distance between the ith sample and the remaining (n_1+n_2-1) samples. Moreover, A is an indicator function. When $l_c^i=l_y^i$, the output is 1. Otherwise, it is 0.

FID: It measures the distance between the generated image and the actual image in feature space

$$\text{FID}(x,y)=||\mu(x)-\mu(y)||^2+\text{Tr}\left(C_x+C_y-2\left(C_xC_y\right)^{1/2}\right) \tag{17}$$

where $\mu(\cdot)$ and $C_\cdot$ are the mean and covariance matrices of the feature vectors, respectively. $\text{Tr}(\cdot)$ denotes the trace of the matrix. The feature vectors of an image are typically derived from the pretrained InceptionV3 [42] model. However, given the distinct imaging mechanisms of natural and SAR images, our approach in practice involves utilizing the SegFormer [43] model, which has been explicitly pretrained on SAR images instead of InceptionV3.

2) Results and Analysis: To validate the performance of style transfer learning, we visualize the qualitative results in Fig. 3. The figure presents raw, generated, and reconstructed images side by side. From an overall point of view, it can be observed that the generated target images grasp the main characteristics of each terrain while differing in detail. For example, while the architectural layout in Naples is very similar to that in Jiujiang, the luminosity features are inherited from Jiujiang. This highlights the ability of the generator to differentiate and extract distinctive feature representations. At the same time, it can also be noticed that transferring the images from Jiujiang to itself can be regarded as an encoding–decoding process. The high similarity of reconstructed images indicates the bidirectionality and stability of the generator.

Table VI shows the quantitative evaluation of real and generated images in different scenes. The results show that the mean and variance of the generated images in each scene are similar to those of the real ones. This indicates that the model can effectively generate target images similar to real images in the spatial domain style. It can be observed that the generated image exhibits a significantly broader dynamic range compared to its real counterpart. Given the limited pretraining data, we conjecture that the model struggles to capture the full extent of the dynamic range inherent in real SAR images. Encouragingly, this limitation does not manifest in any discernible distortion in the image's content or style. The ENL of generated images in most scenes is higher than that of the corresponding real image, showing the generated images less noisy. One possible explanation is that MFSSL does not regard the speckle noise in SAR images as the style and ignores it during the learning. Based on the LS and CS metrics, the brightness and contrast similarity of the generated image closely resemble those of the real image, consistent with the visualization results. Similarly, FID demonstrates remarkable similarity in the feature space. This is attributed

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TABLE VI
QUANTITATIVE EVALUATION OF GENERATED IMAGES ON FOUR SCENES

Area		Mean	Variance	DR	ENL	LS↑	CS↑	1NN*	FID↓
Napoli	real	64.101	1434.652	47.092	5.931	0.900	0.880	0.759	0.945
	generated	64.957	1425.224	149.133	5.980				
PoDelta	real	121.210	1988.219	48.930	8.722	0.936	0.922	0.744	0.979
	generated	121.363	1934.078	76.417	8.850				
Rosenheim	real	119.402	2178.605	26.459	10.242	0.931	0.881	0.682	1.252
	generated	118.815	2086.694	45.525	10.427				
JiuJiang	real	121.115	3284.395	63.142	5.868	0.918	0.918	0.712	0.552
	generated	119.319	2976.086	164,125	6.159				

1 : ↑ means the larger the value, the higher the similarity.
2 : ↓ means the lower the value, the higher the similarity.
3 : * represents the minimal distributional variance when the value is 0.5.

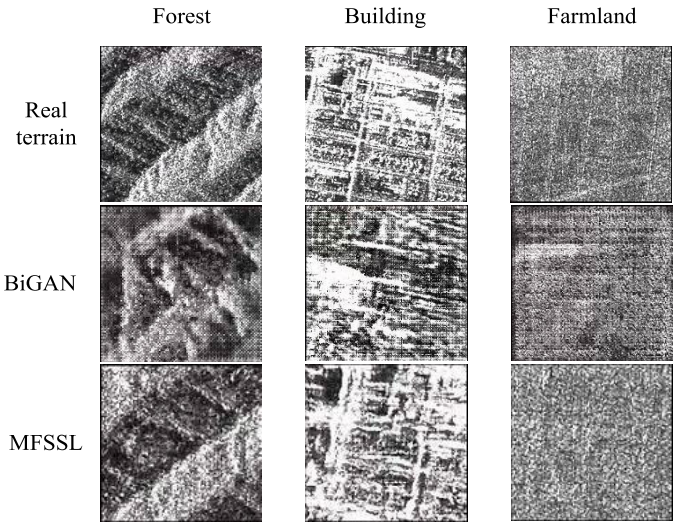


Fig. 4. SAR images generated by BiGAN and MFSSL.

to incorporating constraints on style feature similarity in the upstream task. The accuracy of the 1-NN classifier shows that the multidomain style transfer process is not a simple memory, which can effectively separate the image's content and style information and generate the corresponding target style image. This shows that MFSSL can cover the style differences of images under different image factors. It learns the intrinsical feature among various terrain categories of SAR images, which ensures the feature abstraction capability of the generator across scenes.

In addition, Fig. 4 shows the SAR images generated by BiGAN [44] only with one image factor (statistical distribution and generalized gamma distribution [45]) and our proposed MFSSL considering multi-image factors. It can be easily observed that the SAR images generated by MFSSL are more transparent and plausible, which shows that the factor-oriented generative model has tremendous advantages over direction-less feature learning in complex SAR images.

C. Terrain Classification in the Downstream Task

1) Settings and Metrics: In the downstream terrain classification task, the encoder E pretrained on the upstream task is transferred, and the classifier head is added for the complete terrain classification pipeline. The classifier structure is shown in Table VII, whose last layer uses the convolutional layer instead of the fully connected layer to output the category probability and reduce the number of weights. For fine-tuning, 32 samples of each terrain are randomly selected from the patch-level training dataset in the upstream task. In this stage, the batch size is set to 8, the training duration spans 200 epochs, the initial learning rate is 0.0001, and the learning rate starts to decay after 100 epochs. The Adam optimizer with $\beta_1 = 0$ and $\beta_2 = 0.999$ is used to fine-tune the weights of all models in this stage.

TABLE VII
ARCHITECTURE OF THE CLASSIFIER HEAD

Layer	Norm	Actvation	Output shape
Feature	-	-	$12 \times 12 \times 512$
Conv4×4	BN	ReLU	$6 \times 6 \times 256$
Conv4×4	BN	ReLU	$3 \times 3 \times 128$
Conv4×4	BN	ReLU	$1 \times 1 \times 64$
Conv1×1	-	-	$1 \times 1 \times 4$

*Number of parameters : 2.753×10^6.

The test dataset is constructed by overlapping cropping the broad SAR scenes. The size of each block is 200×200, and the slide is 86. Since the central region with rich context can be inferred with high confidence, the patch-level predicted results are obtained through dilation prediction [46]. Specifically, only the prediction of the center region of the block fed to the classifier is selected to form the final prediction. The specific determined process of the size of the central region is given in Section III-C2. Finally, the terrain prediction of the complete SAR scene is obtained by splicing the center-predicted regions of all test blocks.

Notably, the predicted result of the classifier is the terrain category of the whole block, not each pixel in this block. Compared with pixel-level label data, segmenting terrain directly from patch-level samples with less supervised information is challenging. However, the drop in the cost of time and labeling is attractive. In our experiments, patch-level

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prediction performance is comparable to pixel-level prediction (DeeplabV3+) on the limited labeled samples described at length in Section III-C3.

In the downstream terrain classification task, overall accuracy (OA), Kappa coefficient, mean intersection over union (MIoU), and frequency weight intersection over union (FWIoU) are used to measure the overall classification performance. The F1 score is selected to measure the classification performance of each category. For the convenience of representation, the total number of pixels whose true category j is identified as a category i is denoted as p_{ij}, the total number of samples is denoted as N, and the number of categories is denoted as K. The above metrics are defined as follows.

OA: It is the percentage of accurately classified pixels out of all pixels

$$\mathrm{OA} = \frac{\sum_{i=0}^{K} p_{ii}}{\sum_{i=0}^{K}\sum_{j=0}^{K} p_{ij}}. \tag{18}$$

Kappa: It is an indicator for consistency checks and can also be used to measure the effect of classification

$$\mathrm{Kappa} = \frac{\mathrm{OA} - P_e}{1 - P_e},\ P_e = \frac{1}{N^2}\sum_{i=1}^{K}\left(\sum_{j=0}^{K} p_{ij}\right) \times \left(\sum_{j=0}^{K} p_{ji}\right) \tag{19}$$

where P_e is the hypothetical probability of chance agreement.

MIoU: It calculates the ratio of the intersection and union of two sets of true and predicted values

$$\mathrm{MIoU} = \frac{1}{K+1}\sum_{i=0}^{K} \frac{p_{ii}}{\sum_{j=0}^{K} p_{ij} + \sum_{j=0}^{K} p_{ji} - p_{ii}}. \tag{20}$$

FWIoU: It is an improvement of MIoU, which sets weights according to the frequency of occurrence of categories

$$\mathrm{FWIoU} = \frac{1}{\sum_{i=0}^{K}\sum_{j=0}^{K} p_{ij}} \times \sum_{i=0}^{K} \frac{p_{ii}\sum_{j=0}^{K} p_{ij}}{\sum_{j=0}^{K} p_{ij} + \sum_{j=0}^{K} p_{ji} - p_{ii}}. \tag{21}$$

F1 score: It is the harmonic mean of recall and precision

$$\mathrm{Precision} = \frac{p_{ii}}{\sum_{j=0}^{K} p_{ij}} \tag{22}$$

$$\mathrm{Recall} = \frac{p_{ii}}{\sum_{j=0}^{K} p_{ji}} \tag{23}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{Precison} \times \mathrm{Recall}}{\mathrm{Precison} + \mathrm{Recall}}. \tag{24}$$

2) Hyperparameters Determination: In this section, three crucial hyperparameters are determined by the experiments. They are the size of the image center region for prediction, the size of the image block, and the number of samples.

a) Size of the center prediction region: The size of the center region in the test block is an important factor that affects the classification fineness and the smoothness of the boundary between blocks. Large central regions have high efficiency and low classification accuracy, and small central regions are the opposite. This exploration is conducted with 16 samples per category, and the patch size is 128 × 128. Table VIII shows the classification performance and test time in the Jiujiang data with different sizes of the central prediction region. There is a slight improvement in classification performance as the size of the central region decreases, but the inference time climbs rapidly. When the size of the central region is smaller than 28 × 28, the classification accuracy is slightly improved, and the time required is almost doubled. Considering both performance and efficiency, we set the size of the central region to 28 × 28 in the following experiments.

TABLE VIII
ACCURACIES AND SPEED OF DIFFERENT SIZES OF THE CENTRAL AREA IN JIUJIANG

Size	OA	kappa	MIoU	FWIoU	Times
20×20	**0.757**	**0.645**	**0.514**	**0.687**	1145 s
28×28	**0.757**	**0.645**	**0.514**	**0.687**	609 s
48×48	0.755	0.643	0.512	0.685	204 s
68×68	0.741	0.624	0.499	0.669	109 s
88×88	0.744	0.628	0.502	0.674	62 s
108×108	0.741	0.624	0.499	0.669	42 s
128×128	0.744	0.628	0.502	0.674	30 s

TABLE IX
ACCURACIES WITH DIFFERENT SIZES OF THE BLOCK IN JIUJIANG

Size	OA	kappa	MIoU	FWIoU
50×50	0.765	0.655	0.527	0.693
70×70	0.753	0.639	0.514	0.678
100×100	0.783	0.678	0.544	0.709
128×128	0.771	0.668	0.541	0.706
150×150	**0.802**	**0.705**	0.569	**0.732**
200×200	0.794	0.698	**0.573**	0.724
250×250	0.721	0.598	0.457	0.647
300×300	0.673	0.545	0.426	0.604

TABLE X
OA WITH DIFFERENT NUMBERS OF SAMPLES IN JIUJIANG

Number of samples per category	OA (with pre-training)	OA (without pre-training)
8	**0.731**	0.366
16	**0.725**	0.490
32	**0.710**	0.696
40	**0.781**	0.731
50	0.787	**0.803**
64	0.779	**0.850**
128	0.786	**0.853**

b) Size of image blocks: The prediction of the terrain category in the central region relies on the contextual information provided by the cropped image block. Too small blocks provide limited contextual information, while too large blocks bring information redundancy, hindering classification performance. The experiment is carried out with 32 training samples of each terrain and center prediction region with a

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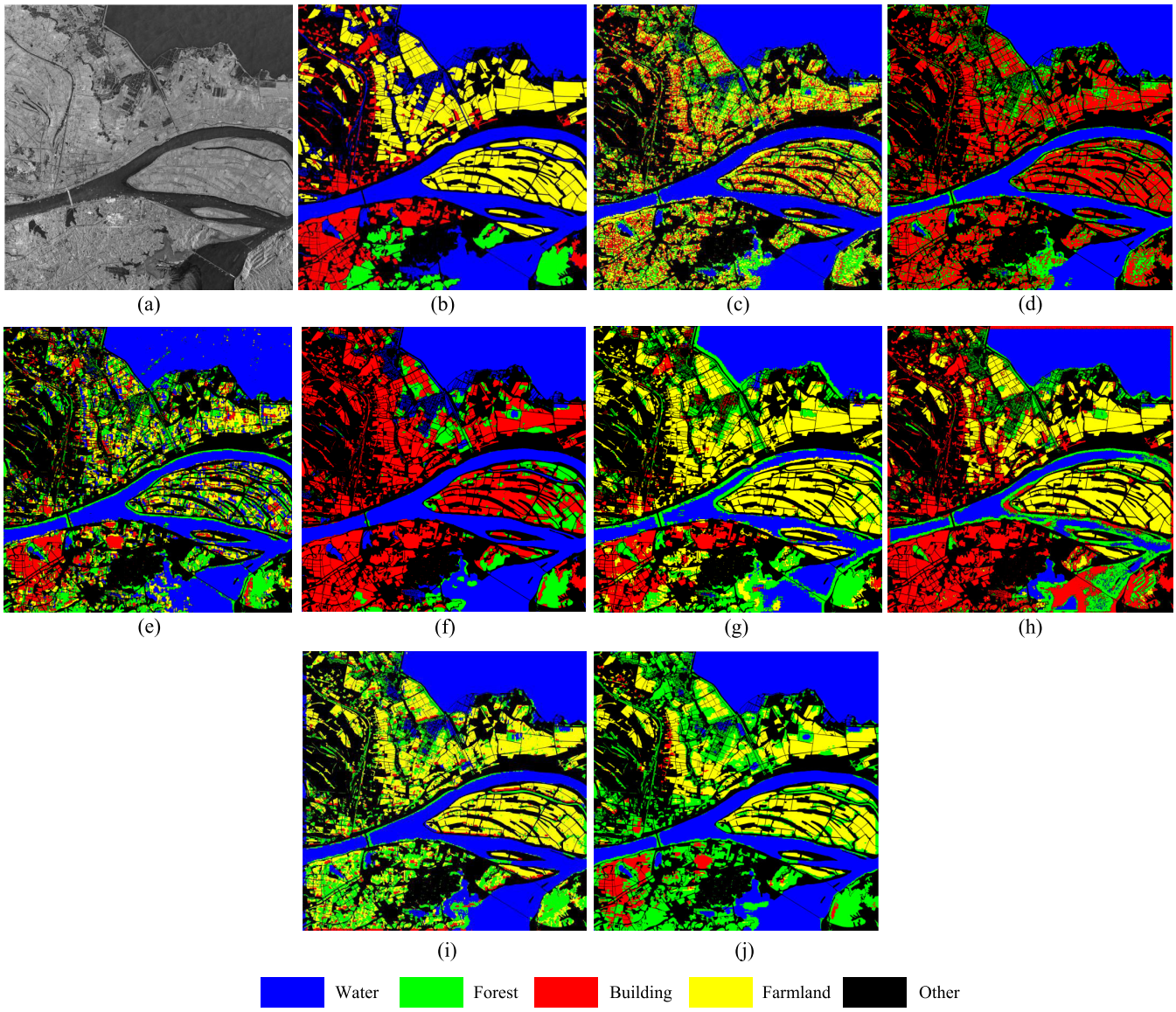


Fig. 5. Visual predicted results of different methods in Jiujiang. (a) SAR image. (b) Ground truth. (c) ResNet-18. (d) SAR-CNN. (e) Deeplabv3+. (f) SegFormer. (g) MAE. (h) CCNR. (i) Baseline. (j) MFSSL.

size of 28 × 28 pixels. Table IX shows the classification performance on different sizes of blocks in Jiujiang. The result shows that at 150 × 150 and 200 × 200, the model achieves more performance than others. Moreover, $MIoU$ in size of 150 × 150 is inferior to 200 × 200. The reason is its insufficiency in the perception of the sparse category (forest). In the following experiments, we set the size of the block to 200 × 200 to ensure the recognition ability of rare classes and the efficiency of the test process.

c) Number of samples: In general, supervised methods outperform SSL methods given sufficient labeled data, but well-labeled data are scarce, especially in SAR classification. Thus, the advantage range of the proposed MFSSL with small amounts of label data is explored here. With a block size of 200 × 200 and a central prediction region of 28 × 28, the experimental results of different numbers of training samples are illustrated in Table X. The model, without inheriting weights of the encoder in the upstream task, is trained with the same labeled samples from scratch and can be regarded as supervised learning. The results show that MFSSL performs better than the supervised methods when the number of training samples is less than 40 in Jiujiang. In the following experiments, we set the number of samples in each category to 32 to explore the potential of models under the common scenario of limited labeled data in real applications.

3) Experiments in Broad Scenes: Fig. 5(a) and (b) shows the processed SAR image of Jiujiang and its corresponding ground truth. Fig. 6(a) and (b), Fig. 7(a) and (b) also show the SAR image and the corresponding ground truth, respectively. The proposed method is compared with six approaches: Resnet-18 [47], SAR-CNN [48], deeplabv3+ [49], SegFormer [43], MAE [50], and CCNR [51], in three broad scenes to demonstrate the effectiveness further. The model P is regarded as the baseline without using the pretrained weights. As a classic classification network, the number of weights in Resnet18 is approximately equivalent to that of P. SAR-CNN is a two-stage method based on a CNN architecture that uses a distance metric for SAR image classification in the second

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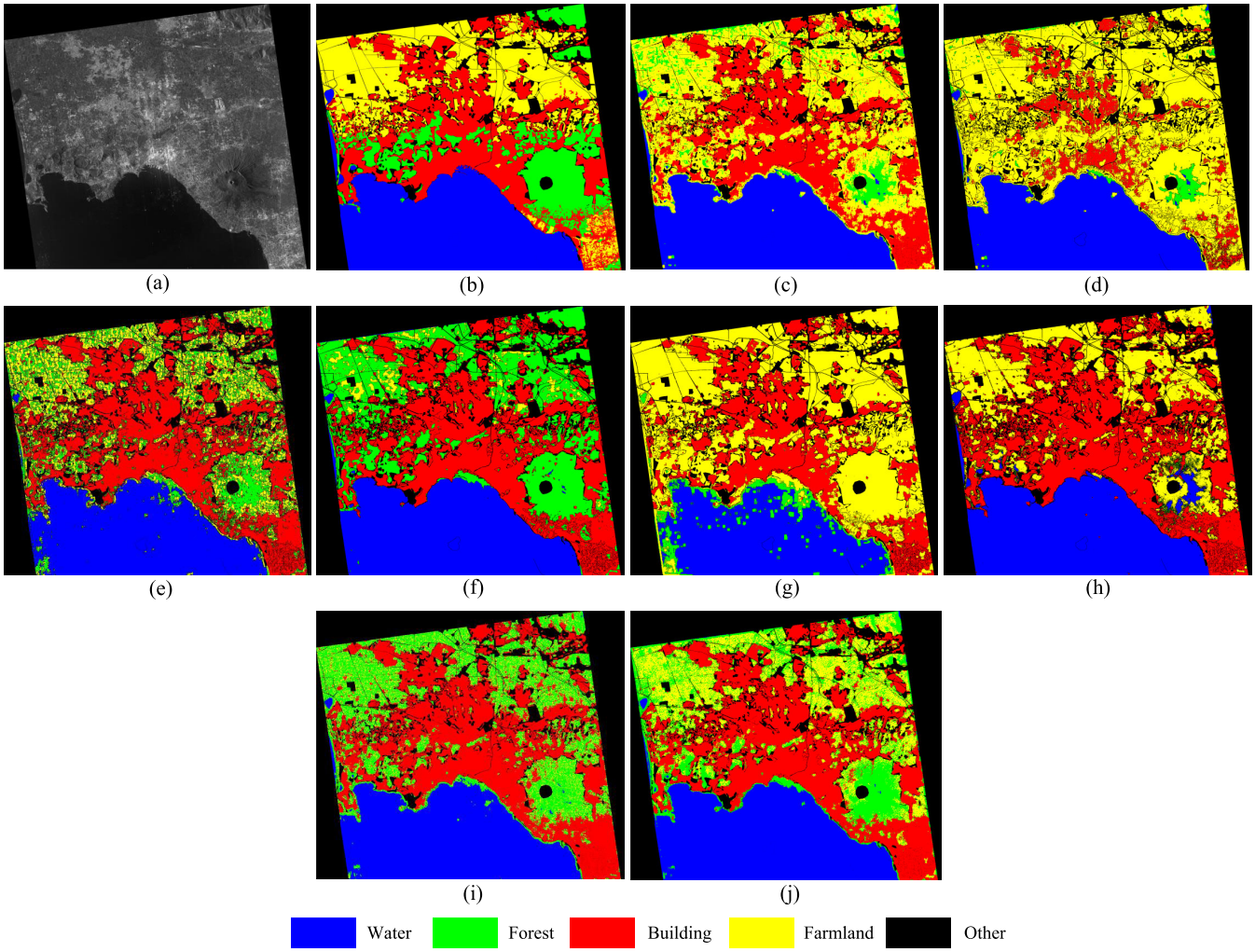


Fig. 6. Visual predicted results of different methods in Napoli. (a) SAR image. (b) Ground truth. (c) ResNet-18. (d) SAR-CNN. (e) Deeplabv3+. (f) SegFormer. (g) MAE. (h) CCNR. (i) Baseline. (j) MFSSL.

stage. For this method, we use ResNet-18 as the classification network in the first stage so that it has a set of weight parameters close to P. Deeplabv3+ and SegFormer, as mainstream segmentation networks, have considerable performance in multiple segmentation tasks using CNN and Transformer structures, respectively. In the experiment, the backbone of Deeplabv3+ is ResNet-50. MAE is an advanced generative SSL method. CCNR is an SAR image terrain classification algorithm using cross-region context and noise regularization. Based on the hyperparameter experiments in III-C2, the sizes of blocks and central regions are 200×200 and 28×28, respectively, and the number of samples in each category is 32, a total of 128. All the weights in the above models are fine-tuned unless otherwise specified. For a fair comparison, the BN layer in the backbone of the comparative method is replaced with IN.

a) Results of Jiujiang: Fig. 5 shows the classification visual results of different methods in Jiujiang. Fig. 5(a) and (b) shows the real SAR image and its ground truth. In the sight of texture, distinct characteristic differences exist among the various terrain categories. When considering pixel intensity, the visual perception of the remaining categories appears similar except for water. This is why other comparison methods' comparison causes considerable confusion. However, after SSL pretraining, the proposed method and MAE can effectively reduce confusion and appear to be a smoother and more accurate predicted map.

The quantitative results of Jiujiang are listed in Table XI, including OA, kappa, MIoU, FWIoU, and F1 scores of each class. MFSSL achieves the highest accuracy over overall indicators except MIoU. Due to the self-attention mechanism of the transformers, MAE achieves higher accuracy in building and MIoU, and SegFormer achieves higher accuracy in water and forest. Moreover, the cross-regional mechanism allows CCNR to have the highest accuracy on farmland.

b) Results of Napoli: Fig. 6 shows the classification visual results of different methods in Napoli. Fig. 6(a) and (b) shows that the contrast and brightness of farmland and forest are very similar. Therefore, the ability to discriminate them is the key to the performance in this scene. Many isolated pixels are misclassified in Fig. 6(d). This is because the large receptive field reduces the feature difference between forest and farmland in Deeplabv3+ and SegFormer. Naturally, the perception ability drops sharply. While CCNR incorporates

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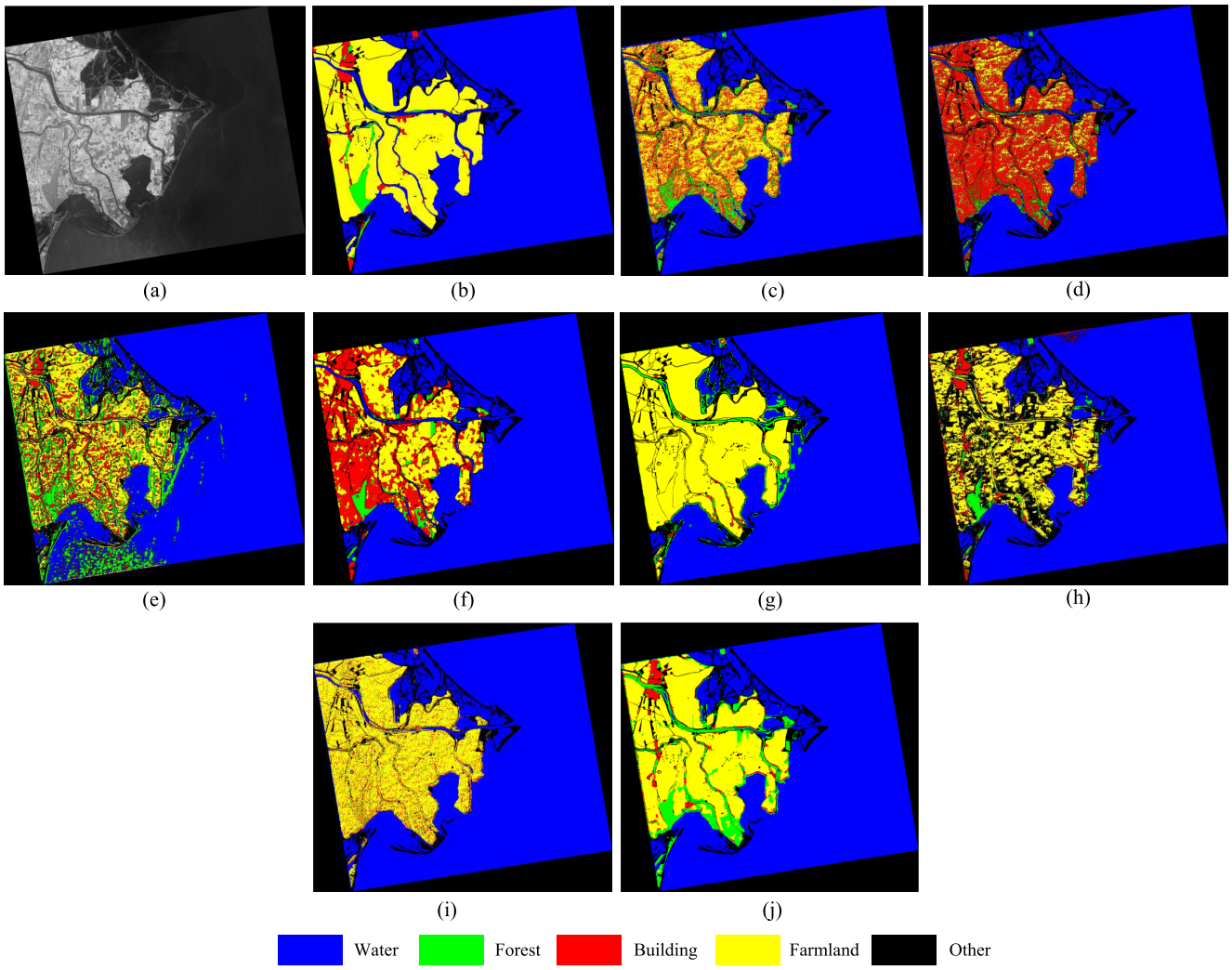


Fig. 7. Visual predicted results of different methods in PoDelta. (a) SAR image. (b) Ground truth. (c) ResNet-18. (d) SAR-CNN. (e) Deeplabv3+. (f) SegFormer. (g) MAE. (h) CCNR. (i) Baseline. (j) MFSSL.

extensive receptive fields, noise regularization distinctly diminishes the count of erroneously classified points, offering a certain degree of alleviation in the misclassification of farmland. Compared with Deeplabv3+ and Segformer, Baseline, ResNet-18, and SAR-CNN use a single scale to avoid large amounts of misclassified pixels. However, due to the limited training samples, these two models tend to classify visually similar regions, such as forests and farmlands, into the same class. Even with SSL, MAE can still not identify farmland and forests effectively. Unlike the above methods, the pretraining and the additional domain information alleviate this, and the perception of farmland is more accurate by MFSSL.

Four overall quantitative indices and F1 scores for evaluating the terrain classification results of Napoli are listed in Table XII. As can be seen from the metrics, SAR-CNN has the lowest performance among the compared methods, which is caused by the biased computation of the prototype for terrain with a small number of training samples. The baseline results are subpar, attributed to the challenge of distinguishing between different terrains with similar appearances when utilizing low-information patch-level samples. ResNet-18 has fewer weights than the baseline, but it has a stronger ability to recognize farmland and building regions. Although SegFormer and Deeplabv3+ have an improvement of 1%–3% in the four overall indicators compared with the baseline, the improvement brought about serious confusion between forest and farmland, and features learned from severe overfitting are not credible. In addition, the perception ability of Deeplabv3+ on recognizable waters is not satisfactory. With the pretrained process, it can be noticed that MFSSL has the best performance. With the injection of pretraining information, the ability of the model to distinguish forests significantly improves at the expense of a small amount of farmland classification accuracy. The above experimental results and analysis demonstrate the efficacy of MFSSL under limited labeled data in Napoli. Moreover, it enhances the discernment of terrain features, particularly in distinguishing between farmland and forest, which is achieved by SSL in the upstream task.

c) Results of PoDelta: Fig. 7 shows the classification visual results of different methods in PoDelta. Unlike the appearance of terrain in Napoli, the features of farmland and forest are significantly different. However, the differences between forest and building regions are blurred, which

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TABLE XI
CLASSIFICATION RESULTS OF DIFFERENT METHODS IN JIUJIANG

Method	F1 Score				OA	kappa	MIoU	FWIoU
	Water	Forest	Building	Farmland				
ResNet-18	0.909	0.217	0.337	0.504	0.618	0.454	0.374	0.540
SAR-CNN	0.872	0.163	0.402	0.017	0.508	0.343	0.280	0.413
Deeplabv3+	0.826	0.269	0.476	0.505	0.614	0.440	0.377	0.497
SegFormer	**0.960**	**0.342**	0.496	0.001	0.617	0.468	0.365	0.501
MAE	0.803	0.302	**0.670**	0.794	0.713	0.592	**0.502**	0.610
CCNR	0.699	0.092	0.607	**0.851**	0.646	0.524	0.440	0.554
Baseline	0.906	0.275	0.150	0.747	0.696	0.552	0.416	0.604
MFSSL	0.916	0.315	0.432	0.788	**0.719**	**0.603**	0.490	**0.659**

TABLE XII
CLASSIFICATION RESULTS OF DIFFERENT METHODS IN NAPOLI

Method	F1 Score				OA	kappa	MIoU	FWIoU
	Water	Forest	Building	Farmland				
ResNet-18	0.968	0.236	**0.845**	0.654	0.756	0.665	0.572	0.636
SAR-CNN	0.967	0.159	0.538	0.586	0.646	0.517	0.451	0.503
deeplabv3+	0.943	0.374	0.817	0.545	0.721	0.620	0.547	0.598
SegFormer	**0.978**	**0.496**	0.840	0.219	0.712	0.615	0.533	0.578
MAE	0.869	0.001	0.827	0.651	0.693	0.579	0.489	0.558
CCNR	0.970	0.108	0.813	**0.681**	**0.765**	0.673	0.550	0.620
Baseline	0.968	0.400	0.808	0.307	0.688	0.579	0.512	0.562
MFSSL	0.972	0.454	0.841	0.594	0.761	**0.675**	**0.596**	**0.645**

TABLE XIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS IN PODELTA

Method	F1 Score				OA	kappa	MIoU	FWIoU
	Water	Forest	Building	Farmland				
ResNet-18	0.980	0.208	0.107	0.669	0.832	0.653	0.409	0.806
SAR-CNN	0.980	0.172	0.095	0.426	0.763	0.531	0.344	0.739
Deeplabv3+	0.930	0.110	0.143	0.704	0.785	0.591	0.387	0.754
SegFormer	**0.988**	0.374	0.173	0.680	0.847	0.682	0.454	0.821
MAE	0.968	0.033	0.071	**0.930**	**0.922**	**0.834**	0.465	0.893
CCNR	0.975	**0.496**	0.419	0.746	0.860	0.712	0.535	0.831
Baseline	0.981	0.113	0.097	0.834	0.899	0.780	0.447	0.867
MFSSL	0.979	0.243	**0.458**	0.892	0.918	0.825	**0.550**	**0.895**

challenges identifying them in PoDelta. In line with the performance exhibited by Deeplabv3+ in Napoli, the predicted maps of ResNet-18, SAR-CNN, Deeplabv3+, SegFormer, CCNR, and the baseline all display numerous instances of isolated misclassified pixels. At the same time, MFSSL performs superior visual effects, and the isolated misclassified pixels are much fewer in the classification maps. Nonetheless, when it comes to recognizing delicate water tributaries, the performance of MFSSL and MAE is not pleasing. This discrepancy may be attributed to the pretraining information directing the model toward easily discernible water regions and other intricate terrains. Consequently, MFSSL and MAE overlook the distinctive features of subtle targets within patch-level blocks while exhibiting enhanced accuracy in classifying terrain with large areas. Due to the absence of domain information guidance during the pretraining phase, MAE exhibits reduced sensitivity to sparse terrain within a large scene.

The quantitative results are presented in Table XIII. It can be noticed that MAE has the highest accuracy in OA, kappa, and farmland because of its neglect of rare terrain. MFSSL improves the perception of sparse terrain and has higher accuracy on MIoU, FWIoU, and buildings. CCNR has a stronger recognition ability on forests. Baseline, SAR-CNN, SegFormer, and ResNet-18 yield much better classification

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results than MFSSL in water. These findings align with the analysis of the visual results, providing further support for the observed trends. This demonstrates the capability of our proposed MFSSL to improve feature discrimination, thereby contributing to terrain classification in various scenes. In addition, MAE demonstrates superior accuracy across specific metrics in diverse scenes compared to MFSSL. Nevertheless, MFSSL maintains an edge in its capacity to generalize across multiple scenes.

D. Ablation Study

In this section, we conduct ablation experiments with multifactor and two-loss constraints to investigate their impact on terrain classification within MFSSL further.

1) Multifactor Experiments: In this experiment, we examined the impact of various image factors on classification performance, employing category constraints as the underlying domain labels. As shown in Table I, the association between band and polarization and sensor and resolution is consistent in the utilized dataset. Therefore, one of these factors was considered separately to simplify the experiment. The MIoU of the multifactors ablation experiments in Jiujiang is shown in Table XIV.

The model's classification performance experiences a significant improvement when incorporating the imaging region, band, and sensor factors solely. The terrain distribution information affected by the imaging region is a more robust guide to the model than hue information influenced by the band. The model incorporates all the mentioned multifactor information and further improves classification performance. Incorporating guidance from multiple factors can enhance the model's ability to discern features. Moreover, a more extensive guidance set from multiple factors can further bolster discriminative performance.

2) Feature and Distribution Similarity Loss Experiments: An ablation study is carried out to investigate the effectiveness of feature similarity and distribution similarity constraints. The prediction is in three broad scenes, whose results are shown in Table XV.

The base model contains a classification constraint, an adversarial constraint, and a reconstruction constraint, which exhibits superior performance in PoDelta. This is primarily attributed to the distinct boundaries between terrains under these three constraints. It has preliminarily shown the effectiveness of the architecture of MFSSL in extracting discriminative features. By incorporating the feature similarity constraint into the base model, remarkable improvement can be observed across four overall indicators in Napoli and Jiujiang. Notably, the model exhibits a more substantial accuracy enhancement, specifically in Jiujiang. These findings indicate that the introduction of feature similarity constraint facilitates the extraction of crucial terrain features, thereby ensuring content consistency and enhancing the distinction of complex confusion regions. A slight decline in performance in PoDelta can also be seen, which is due to the inclination of the model to the rare terrain and results in a small amount of confusion with other terrain. The application of distribution similarity constraint yields minimal impact on PoDelta and Napoli. This can be attributed to the similarity between the image factors of these scenes and the majority of scenes in the pretraining data. Conversely, substantial improvement is observed in the classification performance in Jiujiang. It is because, compared to other scenes, Jiujiang exhibits apparent differences in the source, the resolution, and the band. The distribution similarity constraint effectively mitigates these domain disparities, enhancing performance significantly. With feature and distribution similarity constraints, the model attains the highest classification performance in Jiujiang. Notably, it achieves an improvement of approximately 5% in OA, and the other metrics witness an enhancement of nearly 6%.

TABLE XIV
ABLATION STUDY OF MULTIFACTOR IN JIUJIANG

Base (category)	others	MIoU
✓	-	0.467
✓	+ regions	0.483
✓	+ band	0.481
✓	+ sensor	0.477
✓	+ all	**0.490**

1 : "+ all" denotes inclusion region, resolution, sensor, band, and polarization.

The results above illustrate that the feature similarity constraint assists in learning discriminative features from intricate regions. Meanwhile, the distribution similarity constraint facilitates data alignment across distinct distributions, reducing domain disparities.

E. Feature Visualization

To further study the effect of feature similarity constraint and distribution similarity constraint on image features, the image features output by the encoder model are visualized. Feature visualization enables the identification of specific areas of interest within a category. Analyzing these visualized features provides insights into the discrimination of image features across various terrains. Fig. 8 shows the feature visualization of SAR images with different constraints imposed. For the activation regions outlined in the red box, the Vanilla model tends to exhibit rounded feature activation regions on building terrains with high gray-scale values, and its capability to capture edges is insufficient. This indicates that using the image features to distinguish different terrains in complex regions is challenging. Under the constraint of feature similarity, the activation region of the image's features exhibits a significant overlap with the actual terrain region. The activation region of the orange box indicates that, under a single feature similarity constraint, the model tends to misclassify high-gray scatter noise as buildings. However, with the distribution similarity constraint, these misclassifications are significantly reduced. This suggests that the style similarity constraint is beneficial for enhancing the robustness of the model under various styles of SAR images.

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TABLE XV
ABLATION STUDY OF DIFFERENT CONSTRAINTS

Base	feature similarity	distribution similarity	Napoli				PoDelta				Jiujiang			
			OA	kappa	MIou	FWIou	OA	kappa	MIou	FWIou	OA	kappa	MIou	FWIou
✓			0.774	0.689	0.592	0.647	**0.921**	**0.832**	0.561	**0.896**	0.669	0.541	0.437	0.606
✓	✓		**0.783**	**0.702**	**0.606**	**0.661**	0.917	0.824	0.564	0.895	0.704	0.582	0.451	0.639
✓		✓	0.762	0.675	0.590	0.643	0.920	0.829	**0.568**	0.895	0.689	0.565	0.472	0.621
✓	✓	✓	0.761	0.676	0.596	0.645	0.918	0.825	0.550	0.895	**0.719**	**0.603**	**0.490**	**0.659**

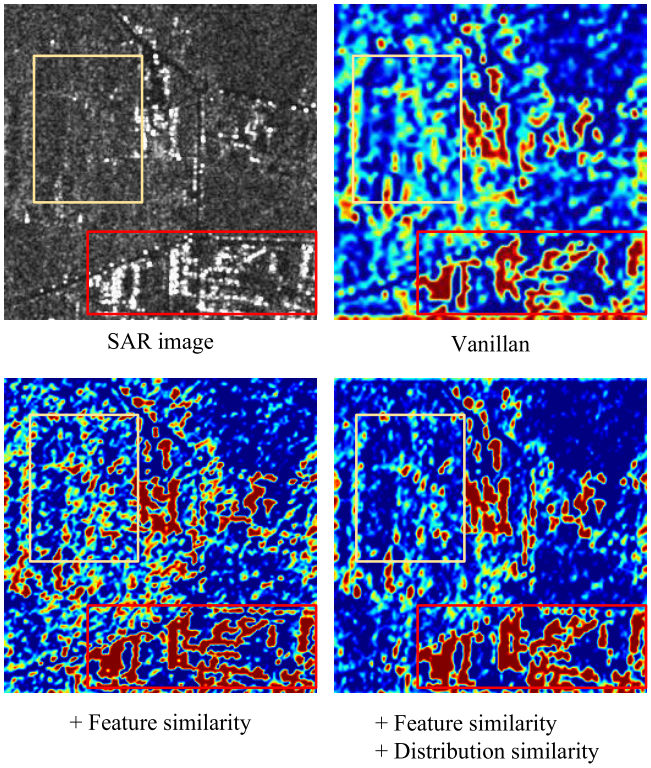


Fig. 8. SAR image feature visualization.

IV. CONCLUSION

This article proposes a self-supervised terrain classification framework called MFSSL to mine the potential from raw data and reduce the reliance on labeled samples. This is achieved mainly by utilizing unlabeled SAR data in an upstream task, which captures various styles of SAR images guided by image factors such as the category, the area, and the radar parameters. The explicit guidance can separate the inherent representation and the manifestation of terrain, thus obtaining quite a good pretrain feature for the downstream task. On this foundation, an excellent performance of terrain classification is acquired with only a few labeled data, and the method is fit for broad scenes. In particular, it is found that the feature similarity constraint and the distribution similarity constraint contribute to the consistency and versatility of image feature representation.

However, the scenes in the experiments are still finite. It is hoped to verify the effectiveness of MFSSL with various scenes, achieve the style transfer across different terrains, and capture a more general feature representation. In addition, it is challenging and attractive to generate terrain images covering new radar factors under the guidance of current radar factors. We also want to use MFSSL on MindSpore, a new deep-learning computing framework.

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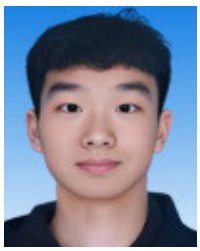
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